

The minimum number of monochromatic five-cycles in a two-coloring of K_9

Xiaojian Zeng

PBC School of Finance, Tsinghua University, 100190

xiaojianzeng@foxmail.com

Abstract

Every red/blue edge-coloring of K_9 contains at least twelve monochromatic five-cycles, counted as ordinary, not necessarily induced, unoriented subgraphs. Equality holds precisely when, after relabeling and possibly exchanging colors, the red graph is $K_5 \sqcup K_4$, with either no edge or one edge added between its components. We give a computer-assisted proof covering all 2^{36} labeled colorings. The certificate consists of 274,668 pairwise nonisomorphic graphs whose exact labeled orbit sizes sum to 2^{36} ; cycle counts are checked by different full-domain implementations. The result concerns this fixed nine-vertex problem. Historical priority remains unverified.

1 Statement and context

For a simple graph G , let $N_5(G)$ be the number of subgraphs isomorphic to C_5 . Thus an injective cyclic vertex sequence is counted once modulo its five rotations and two orientations. Chords are allowed, and distinct cycles on the same five vertices are counted separately. In particular, $N_5(K_5) = 5!/10 = 12$. Identifying a red/blue coloring of K_9 with its red graph gives the objective

$$M(G) = N_5(G) + N_5(\overline{G}), \quad m(C_5) = \min_{|V(G)|=9} M(G).$$

Throughout, equivalence of colorings allows both vertex permutations and exchange of the two colors, the latter corresponding to complementation.

Theorem 1. *For every simple graph G on nine vertices, $M(G) \geq 12$. Equality holds if and only if, up to isomorphism and complementation, G is one of the following two graphs:*

1. $K_5 \sqcup K_4$;
2. $K_5 \sqcup K_4$ with exactly one edge added between the components.

Consequently, $m(C_5) = 12$, with exactly two minimizing coloring classes.

Conlon, Fox, Sudakov, and Wei [1] study threshold Ramsey multiplicity for odd cycles, prove a lower bound of the form $(ck)^k$, and conjecture $m(C_k) = (k-1)!/2$ for sufficiently large odd k . Huang, Lu, Yang, and Chen [2, Theorem 7] state this formula and the two corresponding equality constructions for odd $k \geq k_0$, with an unspecified threshold. Their displayed theorem does not explicitly cover $k = 5$. The present contribution relative to these statements is a complete finite determination at $k = 5$, including equality and an independently checkable coverage certificate. It does not determine their threshold or establish any other odd-cycle case. Exact enumeration at a fixed graph order, together with a separate verification

and an equality list, also occurs in Zeng’s work on a line-graph signature inequality [3]; this is adjacent computational methodology, not a result about the present cycle count.

For the upper bound, the K_5 component of either construction contributes twelve cycles and K_4 contributes none. The optional cross-edge is a bridge, so it lies on no cycle. The complementary graph is $K_{5,4}$, possibly with one edge deleted, and is bipartite. Both constructions therefore have $M = 12$. Their unordered pairs of color-class edge counts are $\{16, 20\}$ and $\{17, 19\}$, which prove their inequivalence. We next establish the lower bound and the exhaustiveness of this equality list.

2 Counting ordinary five-cycles

Lemma 2. *If A is the adjacency matrix of a finite simple graph and d_v is the degree of vertex v , then*

$$10N_5(G) = \text{tr}(A^5) - 5 \sum_v (d_v - 1)(A^3)_{vv}. \quad (1)$$

Proof. The trace counts rooted directed closed walks of length five. Those with five distinct vertices contribute ten per ordinary five-cycle. Every remaining walk uses three or four vertices and has a unique triangle in its traversed support. It either stays inside that triangle or traverses its three edges together with an edge to a fourth vertex, used once in each direction. This follows also by removing the triangle: the two residual traversals must have even incidence at every vertex. Extra edges of the host graph do not affect this classification.

For each triangle T , there are thirty length-five closed walks staying inside it, since the adjacency eigenvalues of K_3 are $2, -1, -1$. Each edge from T to an outside vertex supplies ten further walks, from the five roots and two directions. Hence the nonsimple contribution is

$$\sum_T \left(30 + 10 \sum_{v \in T} (d_v - 2) \right) = 10 \sum_T \sum_{v \in T} (d_v - 1) = 5 \sum_v (d_v - 1)(A^3)_{vv},$$

because $(A^3)_{vv}$ is twice the number of triangles through v . Subtracting this contribution proves (1). $\square$

The certificate program forms A^2, A^3 by integer matrix multiplication and uses $\text{tr}(A^5) = \sum_{i,j} (A^2)_{ij}(A^3)_{ji}$. It applies (1) separately to both colors, checking that each numerator is nonnegative and divisible by ten. At order nine, these matrix computations fit within signed 32-bit integers; masks, orbit sizes, and aggregate masses use 64-bit integers.

A second full-domain method enumerates all 120 vertex orders for each of the $2^{10} = 1024$ graphs on five vertices, and divides the count of successful cyclic orders by ten. For a nine-vertex graph it sums these lookup values over all $\binom{9}{5} = 126$ five-subsets. Every five-cycle has a unique vertex set, so this counts exactly the same ordinary subgraphs. It compares both color counts in every certificate row, covering 34,608,168 five-subsets, with zero mismatches.

3 Certifying the entire domain

The input is a file of 274,668 nine-vertex graphs in graph6 format, originally produced by `geng -1 9`. The coverage argument below treats this file as an arbitrary proposed list: it does not assume the generator’s completeness or rely on a tabulated number of graph classes.

Assign all vertices color zero. Refine a vertex’s color by its old color together with its number of neighbors in each old color class. Assign new numeric colors by lexicographically sorting the distinct signatures. The old color is part of the signature, so cells only split; stop

graph6	Red edges	Red C_5	Blue C_5	Aut	Orbit size
HIPD y{	19	0	12	144	2520
H?B~vrw	20	0	12	2880	126
H~?GW[N	16	12	0	2880	126
HwCW~?^	17	12	0	144	2520

Table 1: All minimizing graph-isomorphism classes, before identifying complementary graphs. The graph6 strings use the input-file labeling.

when their number no longer increases. Starting from the uniform coloring, each ordered partition is invariant under isomorphism.

Enumerate every vertex order in which the final color cells occupy consecutive blocks in numeric color order, exhausting all permutations inside every block. Encode its adjacency matrix by the integer

$$b = \sum_{0 \leq i < j \leq 8} A_{ij} 2^{j(j-1)/2+i},$$

where the indices refer to the chosen vertex order. Define $c(G)$ to be the least such integer and $a(G)$ the number of orders attaining it.

Lemma 3. *Two graphs G, H on nine vertices are isomorphic if and only if $c(G) = c(H)$. Moreover, $a(G) = |\text{Aut}(G)|$.*

Proof. An isomorphism preserves the ordered invariant cells and transports allowed vertex orders bijectively, preserving their adjacency matrices. It therefore preserves the minimum code. Conversely, equality of two minimum codes gives an isomorphism by matching the vertices in the corresponding positions.

Fix one allowed order attaining $c(G)$. Every automorphism preserves the invariant cells and sends this order to another allowed order with the same matrix. Conversely, matching positions between the fixed order and any order with that matrix gives a unique automorphism. This bijection proves the formula for $a(G)$. $\square$

The program rejects duplicate canonical codes and checks that each $a(G)$ divides $9!$. It records the input and canonical graph6 strings, automorphism order, orbit size, and the two cycle counts in one tab-separated row per graph.

Proposition 4 (Exact computation). *The supplied certificate contains 274,668 distinct canonical codes and satisfies*

$$\sum_{G \text{ in the certificate}} \frac{9!}{a(G)} = 68\,719\,476\,736 = 2^{36}. \quad (2)$$

Its minimum value of $M(G)$ is twelve, attained in precisely the four rows of Table 1.

This finite computation is provided by the programs and data described in Section 4. The canonicalization program examines 13,495,956 vertex orders and evaluates (1) on every row. The independent counts described above agree row by row. These exact executions are the computational part of the proof.

By Lemma 3, the listed graph orbits are disjoint and have sizes $9!/a(G)$, by orbit-stabilizer. They form a subset of all labeled graphs on nine vertices. There are precisely $2^{\binom{9}{2}} = 2^{36}$ such graphs, so (2) proves that the subset is the entire domain. Together with Proposition 4, this proves the universal lower bound.

For an explicit identification, label vertices $0, \dots, 8$. The third row has red cliques on $\{4, 5, 6, 7, 8\}$ and $\{0, 1, 2, 3\}$, with no red cross-edge. The fourth has red cliques on $\{3, 4, 5, 6, 8\}$ and $\{0, 1, 2, 7\}$, with the single red cross-edge $\{7, 8\}$. The first two rows are isomorphic to the complements of the fourth and third, respectively. The equality checker verifies every internal and cross edge in these identifications. Thus the four graph classes give exactly the two coloring classes in Theorem 1, completing its computer-assisted proof. They account for $2(126 + 2520) = 5292$ labeled minimizing colorings.

The threshold terminology can also be justified directly. The lower bound proves $r(C_5) \leq 9$. The coloring with red graph $K_4 \sqcup K_4$ on eight vertices has no monochromatic five-cycle, since its blue graph is bipartite. Hence $r(C_5) = 9$.

4 Reproduction and limitations

The accompanying source-and-data supplement contains `evidence/all-nine-vertex.g6`, the complete `evidence/domain-certificate.tsv`, the C/C++ checkers, and Python validation scripts. In a working copy of the supplement, execute

```
python3 evidence/run_certificate.py
```

with Python 3 and Clang supporting C11 and C++17. The script rebuilds the checkers and regenerates their outputs without network access or a graph-library dependency. The supplement's README gives the exact commands for the additional verifier and equality replay.

The additional verifier uses static cells defined by degree, triangle count, and neighbor-degree histogram; it assigns compatible unused vertices successively and minimizes a big-endian adjacency code. It independently checks all automorphism orders and orbit sizes, using 7,837,816 vertex orders. Its cycle counter follows simple paths r, a, b, c, d, r , with r the least vertex and $a < d$. These conditions select one root and direction of every five-cycle, with no restriction on chords. It agrees with every per-color count, recovers (2) and Table 1, and identifies the equality structures by complete edge checks.

Supplementary controls compare canonicalization with all $9!$ orders on twenty graphs and test 400 relabelings. A Burnside computation over the thirty vertex cycle types in S_9 matches all 37 unlabeled edge strata. For each edge count e , the labeled orbit mass is $\binom{36}{e}$. The weighted first moment is

$$\sum_G \frac{9!}{a(G)} M(G) = 2^{36} \frac{1512}{16} = 6\,493\,990\,551\,552,$$

since K_9 has $\binom{9}{5} \cdot 12 = 1512$ five-cycles and each is monochromatic in a fraction $1/16$ of all colorings. The additional verifier rejects a deleted orbit, a duplicate orbit, a changed cycle count, and a changed automorphism order. Its run with undefined-behavior instrumentation reported no error. These are consistency checks; the domain proof is the disjoint-orbit argument.

All decisive arithmetic is exact. The proof depends on the supplied programs and their execution, including compiler, runtime, and hardware correctness; it has not been formalized in a proof assistant. The different implementations ran on the same machine. No assertion is made for other graph orders or odd cycle lengths.

The literature comparison uses the introductions of the two odd-cycle papers cited above, inspected on 9 September 2026. Older sources have not all been accessed or inspected, and historical overlap and priority remain unverified. No claim of first discovery or previously verified open status is made.

Computational and AI assistance. OpenAI Codex assisted with the mathematical argument, program construction, computational verification, and manuscript preparation. A separate model context constructed the additional verifier and reviewed the mathematics. Such review is not external peer review or formal proof-assistant certification.

References

- [1] D. Conlon, J. Fox, B. Sudakov, and F. Wei, *Threshold Ramsey multiplicity for odd cycles*, arXiv:2108.00987v2 (2021). <https://arxiv.org/abs/2108.00987v2>.
- [2] T. Huang, J. Lu, J. Yang, and Y. Chen, *Threshold Ramsey multiplicity and extremal colorings for odd cycles*, arXiv:2609.07285v1 (2026). <https://arxiv.org/abs/2609.07285v1>.
- [3] Zijian Zeng, *Exact Verification of the Line-Graph Signature Inequality at Order Nine*, SSRN preprint (2026). <https://doi.org/10.2139/ssrn.7386040>.